GulzarArwa-FinalProject

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Data Preparation

```
#Load the package needed for data manipulation and visualization
library(tidyverse)

#Import the RNA-seq count and metadata
counts_df <- read.csv("data/counts.csv")

metadata_df <- read.csv("data/meta_data.csv")

#Examine the first six rows of each dataset
head(counts_df[, 1:6])
```

```
##                   X TCGA.GM.A2DL.01A.11R.A18M.07 TCGA.AC.A2QI.01A.12R.A19W.07
## 1 ENSG00000000003.15                         1262                          2922
## 2  ENSG00000000005.6                          120                             4
## 3 ENSG00000000419.13                         1535                          1779
## 4 ENSG00000000457.14                          885                          2574
## 5 ENSG00000000460.17                          328                           586
## 6 ENSG00000000938.13                          604                           625
##   TCGA.A8.A06R.01A.11R.A00Z.07 TCGA.EW.A1PD.01A.11R.A144.07
## 1                         1287                         5617
## 2                            9                          148
## 3                         2255                         1994
## 4                          654                         1335
## 5                          401                          678
## 6                          556                          703
##   TCGA.AO.A12D.01A.11R.A115.07
## 1                         2830
## 2                            2
## 3                         1372
## 4                         1463
## 5                          475
## 6                          561
```

```
head(metadata_df[, 1:6])
```

```
##                       barcode      patient           sample shortLetterCode
## 1 TCGA-GM-A2DL-01A-11R-A18M-07 TCGA-GM-A2DL TCGA-GM-A2DL-01A              TP
## 2 TCGA-AC-A2QI-01A-12R-A19W-07 TCGA-AC-A2QI TCGA-AC-A2QI-01A              TP
## 3 TCGA-A8-A06R-01A-11R-A00Z-07 TCGA-A8-A06R TCGA-A8-A06R-01A              TP
```

```
## 4 TCGA-EW-A1PD-01A-11R-A144-07 TCGA-EW-A1PD TCGA-EW-A1PD-01A              TP
## 5 TCGA-AO-A12D-01A-11R-A115-07 TCGA-AO-A12D TCGA-AO-A12D-01A              TP
## 6 TCGA-AR-A24N-01A-11R-A169-07 TCGA-AR-A24N TCGA-AR-A24N-01A              TP
##            definition sample_submitter_id
## 1 Primary solid Tumor    TCGA-GM-A2DL-01A
## 2 Primary solid Tumor    TCGA-AC-A2QI-01A
## 3 Primary solid Tumor    TCGA-A8-A06R-01A
## 4 Primary solid Tumor    TCGA-EW-A1PD-01A
## 5 Primary solid Tumor    TCGA-AO-A12D-01A
## 6 Primary solid Tumor    TCGA-AR-A24N-01A
```

```r
# Match metadata to the order of samples in the count matrix
sample_ids <- gsub(".", "-", colnames(counts_df)[-1], fixed = TRUE)

meta_aligned <- metadata_df[
  match(sample_ids, metadata_df$barcode),
]

stopifnot(
  !anyNA(meta_aligned$barcode),
  identical(sample_ids, as.character(meta_aligned$barcode))
)

cat("Matched samples:", length(sample_ids))
```

```
## Matched samples: 1231
```

```r
# Identify ESR1 and PGR using their Ensembl gene IDs
gene_ids <- sub("\\..*$", "", counts_df[[1]])

esr1_row <- which(gene_ids == "ENSG00000091831")
pgr_row  <- which(gene_ids == "ENSG00000082175")

stopifnot(length(esr1_row) == 1, length(pgr_row) == 1)

# Extract raw RNA-seq counts for both genes
esr1_counts <- as.numeric(counts_df[esr1_row, -1])
pgr_counts  <- as.numeric(counts_df[pgr_row, -1])

# Calculate summary statistics
gene_summary <- data.frame(
  Gene = c("ESR1", "PGR"),
  Mean = c(mean(esr1_counts), mean(pgr_counts)),
  Median = c(median(esr1_counts), median(pgr_counts)),
  SD = c(sd(esr1_counts), sd(pgr_counts)),
  Minimum = c(min(esr1_counts), min(pgr_counts)),
  Maximum = c(max(esr1_counts), max(pgr_counts))
)

# Display a formatted table
knitr::kable(
  gene_summary,
  digits = 2,
```

```
  caption = "Summary Statistics for ESR1 and PGR Raw RNA-seq Counts"
)
```

Table 1: Summary Statistics for ESR1 and PGR Raw RNA-seq Counts

Gene	Mean	Median	SD	Minimum	Maximum
ESR1	31442.22	17561	38261.25	14	270977
PGR	8791.82	2382	17529.30	1	345065

Data Preparation Explanation: I imported the RNA-seq count matrix and metadata into R and examined the first six rows of each dataset. This allowed me to verify the structure of the datasets before selecting genes and performing the analysis.

Gene Selection and Summary Statistics Explanation: I selected ESR1 (ENSG00000091831) and PGR (ENSG00000082175), two genes involved in hormone receptor signaling in breast cancer. I calculated the mean, median, standard deviation, minimum, and maximum raw RNA-seq counts for both genes across 1,231 samples. ESR1 had a mean of 31,442.22 and a median of 17,561, while PGR had a mean of 8,791.82 and a median of 2,382. Both genes had means higher than their medians, indicating right-skewed raw count distributions.

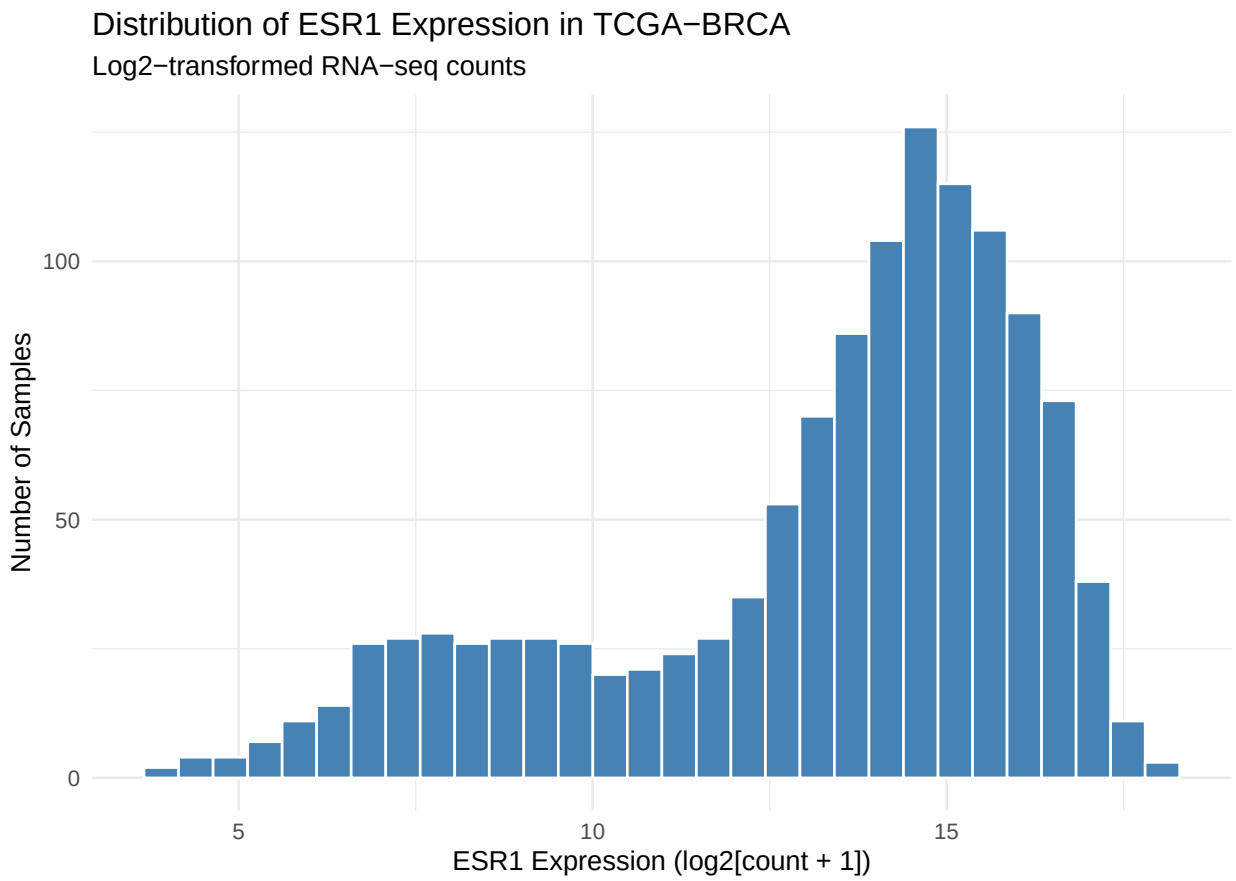
Visualization

Histogram

```
# Log2-transform ESR1 counts
esr1_data <- data.frame(
  ESR1_log2 = log2(esr1_counts + 1)
)

# Create the improved histogram
esr1_histogram <- ggplot(esr1_data, aes(x = ESR1_log2)) +
  geom_histogram(
    bins = 30,
    fill = "steelblue",
    color = "white"
  ) +
  labs(
    title = "Distribution of ESR1 Expression in TCGA-BRCA",
    subtitle = "Log2-transformed RNA-seq counts",
    x = "ESR1 Expression (log2[count + 1])",
    y = "Number of Samples"
  ) +
  theme_minimal()

print(esr1_histogram)
```



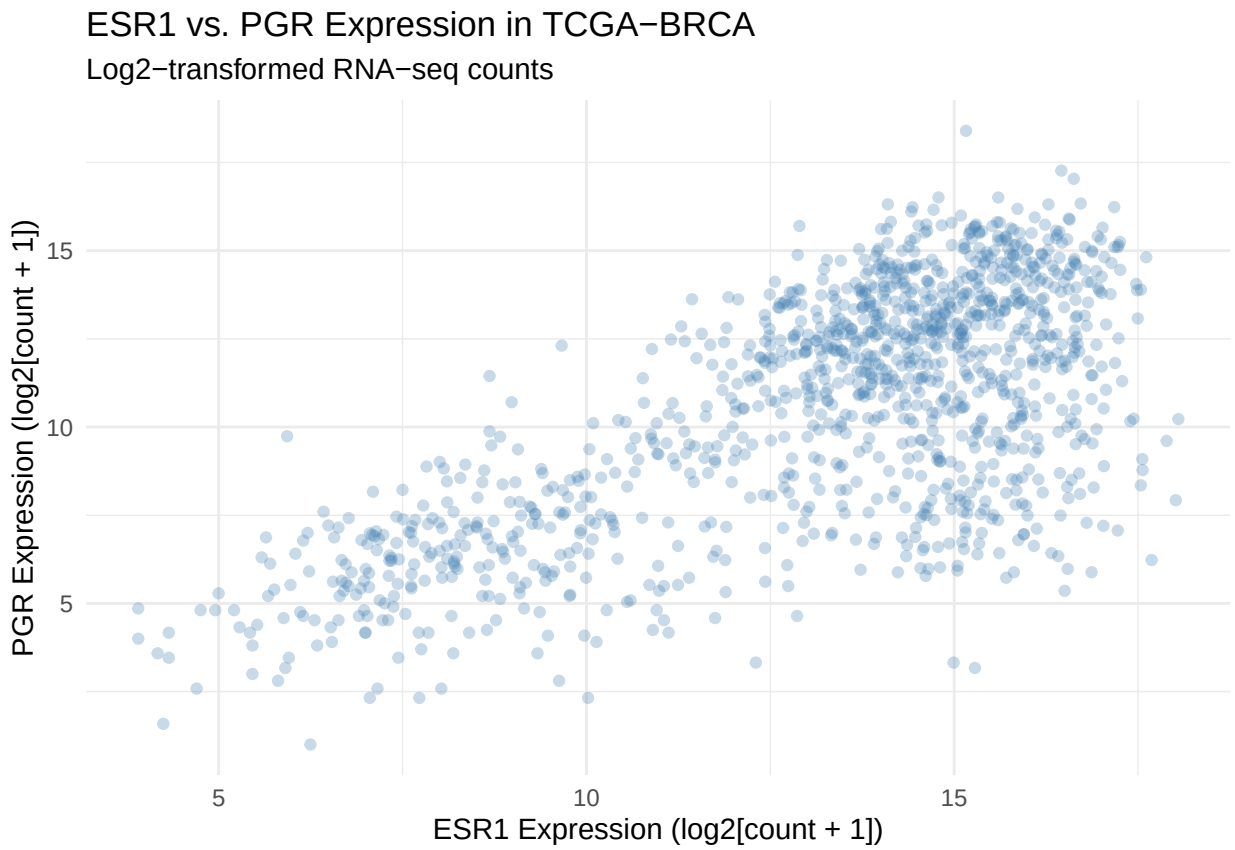
Histogram Explanation: The histogram displays the distribution of ESR1 RNA-seq counts across 1,231 TCGA-BRCA samples after applying a log2(count + 1) transformation. This transformation reduces the visual influence of extremely high counts and makes differences across samples easier to observe. Most samples are concentrated around log2 expression values of 14–16, while another group has lower expression values. The distribution shows substantial variation in ESR1 expression across breast cancer samples.

Scatter Plot

```
# Create a data frame containing ESR1 and PGR counts
scatter_df <- data.frame(
  ESR1_counts = esr1_counts,
  PGR_counts = pgr_counts
)

# Create a scatter plot with log2-transformed counts
scatter_plot <- ggplot(
  scatter_df,
  aes(
    x = log2(ESR1_counts + 1),
    y = log2(PGR_counts + 1)
  )
) +
  geom_point(alpha = 0.3, color = "steelblue") +
  labs(
```

```
    title = "ESR1 vs. PGR Expression in TCGA-BRCA",
    subtitle = "Log2-transformed RNA-seq counts",
    x = "ESR1 Expression (log2[count + 1])",
    y = "PGR Expression (log2[count + 1])"
  ) +
  theme_minimal()
print(scatter_plot)
```



Scatter Plot Explanation: The scatter plot compares ESR1 and PGR RNA-seq counts across TCGA-BRCA samples using log2(count + 1)-transformed values for both genes. The plot shows a general positive association, with samples expressing higher levels of ESR1 often also showing higher PGR counts. However, there is substantial variation, indicating that the relationship is not perfectly linear. Transparency was added to the points to make overlapping observations easier to distinguish.

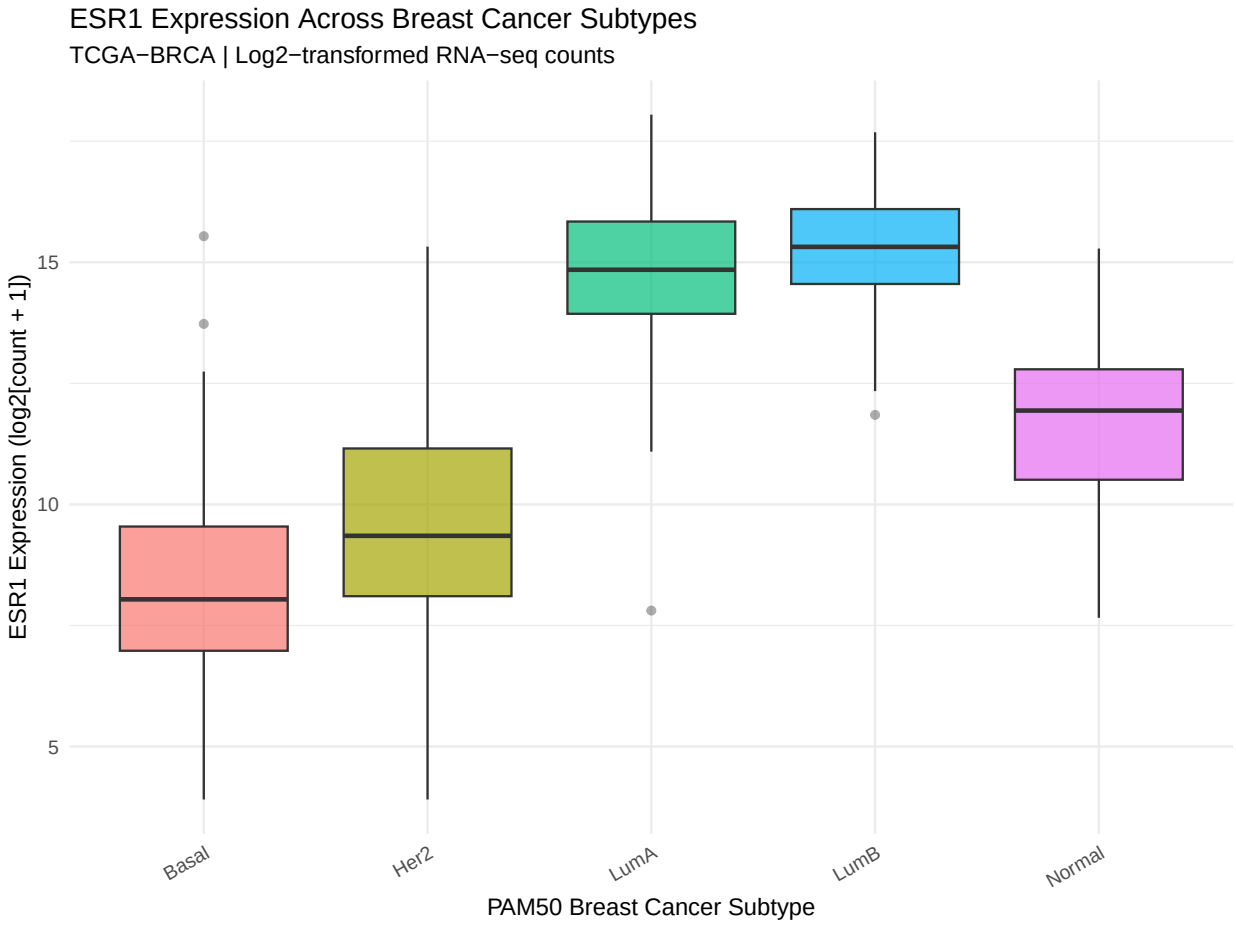
Box Plot

```
# Combine ESR1 expression with correctly matched subtype information
boxplot_data <- data.frame(
  ESR1_log2 = log2(esr1_counts + 1),
  PAM50 = meta_aligned$paper_BRCA_Subtype_PAM50
)

# Remove samples with missing subtype information
boxplot_data <- na.omit(boxplot_data)
```

```r
# Create the box plot
esr1_boxplot <- ggplot(
  boxplot_data,
  aes(x = PAM50, y = ESR1_log2, fill = PAM50)
) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.4) +
  labs(
    title = "ESR1 Expression Across Breast Cancer Subtypes",
    subtitle = "TCGA-BRCA | Log2-transformed RNA-seq counts",
    x = "PAM50 Breast Cancer Subtype",
    y = "ESR1 Expression (log2[count + 1])"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 30, hjust = 1)
  )

print(esr1_boxplot)
```



Box Plot Explanation: The box plot compares log2-transformed ESR1 RNA-seq counts across PAM50 breast cancer subtypes. Luminal A and Luminal B samples generally show higher ESR1 counts, while Basal samples show substantially lower counts. The variation within each subtype highlights differences in ESR1

expression across individual tumor samples.

Heatmap

```r
# Load packages
library(ComplexHeatmap)
library(circlize)

# Select the 10 genes with the highest average raw counts
gene_means <- rowMeans(counts_df[, -1])
top_genes <- counts_df[[1]][order(gene_means, decreasing = TRUE)[1:10]]

# Create a matrix containing the selected genes
heatmap_matrix <- as.matrix(
  counts_df[counts_df[[1]] %in% top_genes, -1]
)

rownames(heatmap_matrix) <- counts_df[[1]][
  counts_df[[1]] %in% top_genes
]

# Log2-transform the counts
heatmap_matrix_log <- log2(heatmap_matrix + 1)

# Prepare the aligned breast cancer subtype annotation
pam50 <- as.character(meta_aligned$paper_BRCA_Subtype_PAM50)
pam50[is.na(pam50) | pam50 == ""] <- "Unknown"

subtype_colors <- c(
  LumA = "#4DAF4A",
  LumB = "#377EB8",
  Basal = "#E41A1C",
  Her2 = "#984EA3",
  Normal = "#FF7F00",
  Unknown = "#BDBDBD"
)

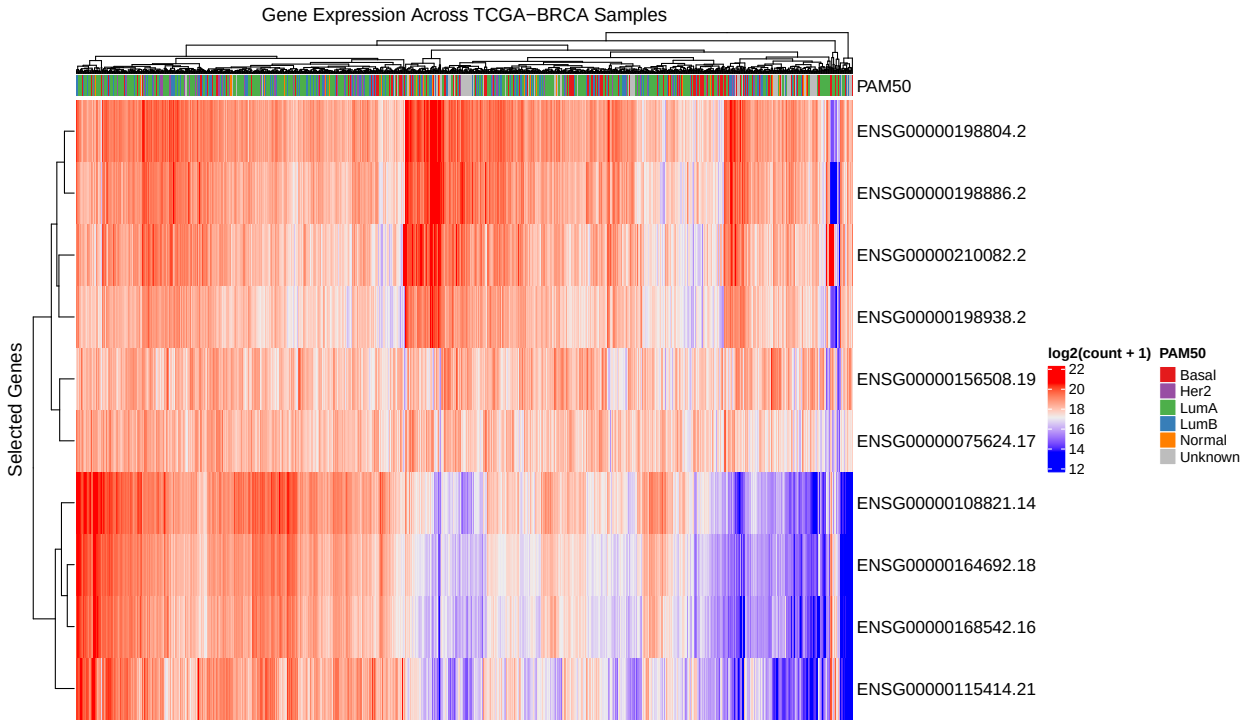
subtype_colors <- subtype_colors[
  names(subtype_colors) %in% unique(pam50)
]

pam50_annotation <- HeatmapAnnotation(
  PAM50 = pam50,
  col = list(PAM50 = subtype_colors)
)

# Create and display the heatmap
final_heatmap <- Heatmap(
  heatmap_matrix_log,
  name = "log2(count + 1)",
  top_annotation = pam50_annotation,
  show_column_names = FALSE,
```

```
  show_row_names = TRUE,
  column_title = "Gene Expression Across TCGA-BRCA Samples",
  row_title = "Selected Genes",
  cluster_columns = TRUE,
  cluster_rows = TRUE
)

draw(final_heatmap)
```



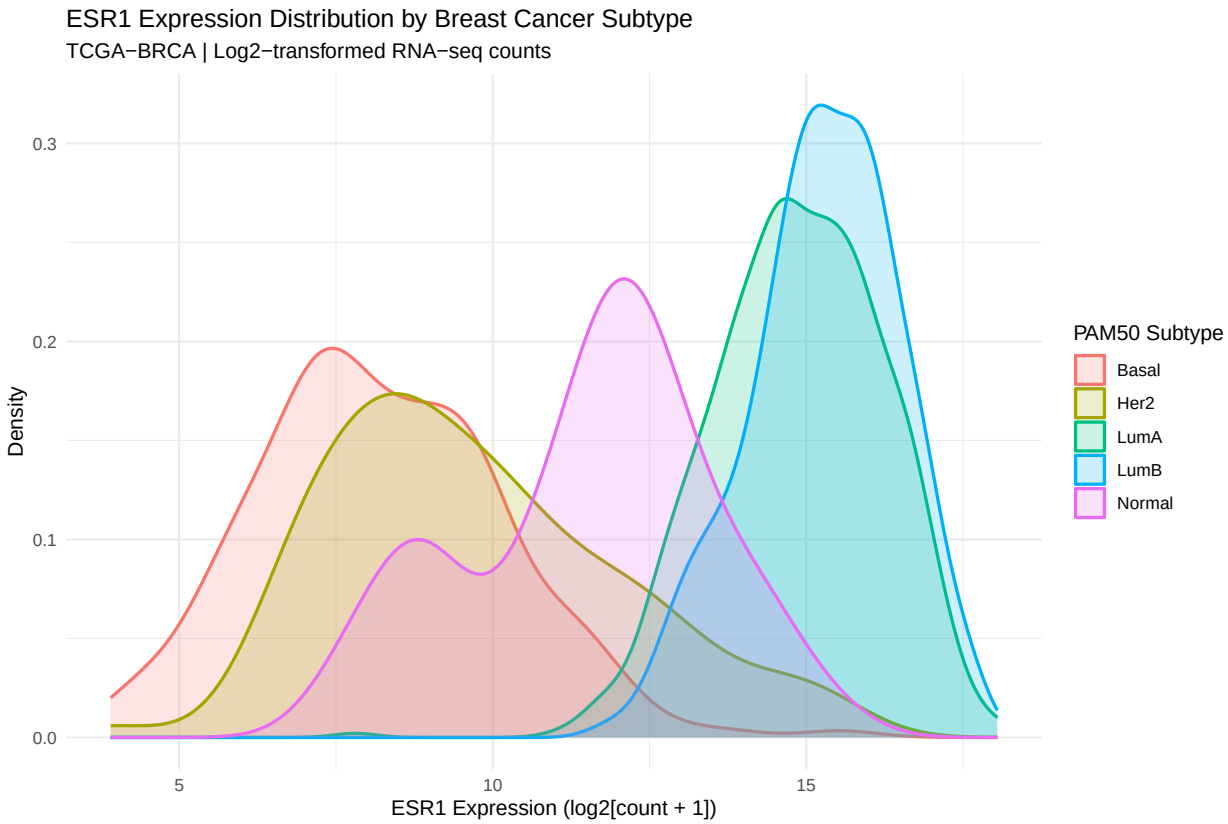
Heatmap Explanation: The heatmap displays log2(count + 1)-transformed RNA-seq counts for the 10 genes with the highest average raw counts across 1,231 TCGA-BRCA samples. Hierarchical clustering groups genes and samples according to similarities in their expression patterns, while the annotation bar identifies each sample's PAM50 breast cancer subtype. The heatmap reveals variation in gene expression across samples, with some groups exhibiting similar expression profiles. The log2 transformation helps make these patterns easier to visualize despite the wide range of raw counts.

Density Plot

```
# Compare ESR1 expression distributions across PAM50 subtypes
esr1_density <- ggplot(
  boxplot_data,
  aes(x = ESR1_log2, fill = PAM50, color = PAM50)
) +
  geom_density(alpha = 0.2, linewidth = 0.8) +
  labs(
    title = "ESR1 Expression Distribution by Breast Cancer Subtype",
    subtitle = "TCGA-BRCA | Log2-transformed RNA-seq counts",
    x = "ESR1 Expression (log2[count + 1])",
```

```
    y = "Density",
    fill = "PAM50 Subtype",
    color = "PAM50 Subtype"
  ) +
  theme_minimal()

print(esr1_density)
```



Density Plot Explanation: The density plot compares the distribution of log2-transformed ESR1 counts across PAM50 breast cancer subtypes. Luminal A and Luminal B samples tend to have higher ESR1 expression, while Basal samples are concentrated at lower expression levels. The overlapping curves also show variation within and between subtypes.